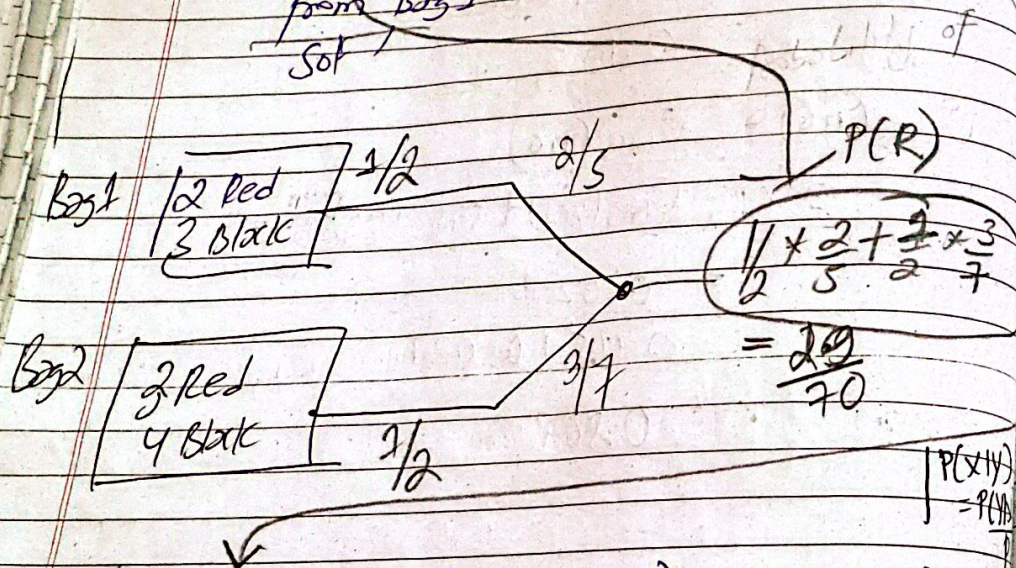


Another

7. Bag 1 contain 2 Red & 3 Black balls. Bag 2 contain 3 Red & 4 Black balls - One ball is drawn at random from one of these bags & is red. find the probability it is drawn from Bag 1.

Sol



$$P(B_1|R) = \frac{P(R|B_1) * P(B_1)}{P(R)}$$

$$= \frac{2}{5} * \frac{1}{2} = 0.4$$

$$\frac{29}{70}$$

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

dataset = Y

features

$$P(Y|X_1, X_2, \dots, X_N)$$

(Assuming, every feature OR independent of each other)

$$= \frac{P(X_1|Y) * P(X_2|Y) * \dots * P(X_N|Y) * P(Y)}{P(X_1) * P(X_2) * \dots * P(X_N)} \quad \text{--- (I)}$$

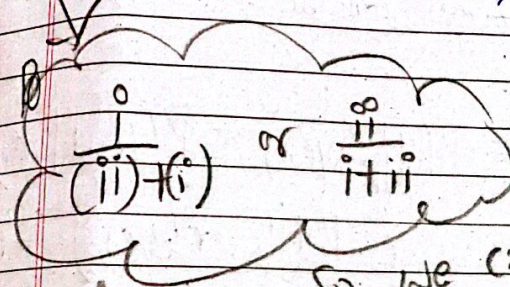
dataset = NO

features

$$P(N|X_1, X_2, \dots, X_N)$$

$$= \frac{P(X_1|N) * P(X_2|N) * \dots * P(X_N|N) * P(N)}{P(X_1) * P(X_2) * \dots * P(X_N)} \quad \text{--- (II)}$$

Imp



$$P(Y|X_1, X_2, \dots, X_N) = P(X_1|Y) * P(X_2|Y) * \dots * P(X_N|Y) * P(Y)$$

Person	x_1 COVID (Y/N)	x_2 Flu (Yes/No)	Fever (Yes/No)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

Ques, Person (flu, covid)

If person has both flu & covid then fever will be?

Ans)

$$P(\text{fever} = \text{yes} | \text{flu, covid} = \text{yes}) = P(\text{flu} | \text{fever} = \text{yes}) * P(\text{covid} | \text{fever} = \text{yes})$$

or

$$P(\text{fever} = \text{No} | \text{flu, covid} = \text{yes}) = P(\text{flu} | \text{No}) * P(\text{covid} | \text{No})$$

So, probability

Eqn (1)

$$P(\text{fever} = \text{yes} | \text{flu, covid}) = \frac{3}{9} * \frac{4}{7} * \frac{7}{10} = 0.121$$

$$P(\text{fever} = \text{No} | \text{flu, covid}) = \frac{1}{3} * \frac{1}{3} * \frac{3}{10}$$

Step 1 - Prior probability:-

$$P(\text{fever} = \text{yes}) = \frac{7}{10}$$

$$P(\text{fever} = \text{No}) = \frac{3}{10}$$

Step 2 - Conditional probability:- fever

	yes	No
covid	5/7	1/3
	5	3/7
		2/3

	yes	No
flu	4/7	1/3
	4	3/7
		2/3
	6	4/7

PCA

principal component

is VS

manre

Pipeline

Hyperparameter

Tuning

Grid Search

Normalized probabilities

Date _____
Page _____

for actual probability values

$$P(\text{fever} = \text{yes} | \text{flu, cold}) = \frac{0.171}{0.171 + 0.033} = 0.839$$

$$P(\text{fever} = \text{no} | \text{flu, cold}) = \frac{0.033}{0.171 + 0.033} = 0.162$$

So, he'll have fever.

NAIVE BAYES CLASSIFIER:-

Date _____
Page _____

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
P4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

(outlook = sunny, Temperature = cool, Humidity = high, Wind = strong). play tennis = ?

playTennis $\rightarrow$ Yes 9
 NO 5

$$P(\text{playTennis} = \text{yes}) = \frac{9}{14} = 0.643$$

$$P(\text{playTennis} = \text{no}) = \frac{5}{14} = 0.357$$

outlook $\neq N$
 sunny
 overcast
 rain

outlook		Yes	No
sunny	5	(2/9)	3/5
overcast	4	4/9	0/5
rain	5	3	2/5

Humidity		Yes	No
high	7	3/9	4/5
normal	7	6/9	1/5

Temperature		Yes	No
Hot	4	2/9	2/5
Mild	6	4/9	2/5
Cool	4	3/9	1/5

Windy

weak	8
strong	6

Yes	No
6/9	2/5
3/9	3/5

$$P(\text{outlook} = \text{sunny} | \text{playTennis} = \text{yes}) = \frac{2}{9}$$

Similarly others.

outlook = sunny, Temperature = cool, Humidity = high,
 Wind = strong

$$v_{NB} = \underset{v_j \in \{\text{yes}, \text{no}\}}{\text{argmax}} P(v_j) \prod_i P(a_i | v_j)$$

$$= \underset{v_j \in \{\text{yes}, \text{no}\}}{\text{argmax}} P(v_j) P(\text{outlook} = \text{sunny} | v_j) \cdot P(\text{Temperature} = \text{cool} | v_j) \cdot P(\text{Humidity} = \text{high} | v_j) \cdot P(\text{Wind} = \text{strong} | v_j)$$

→ $P(\text{yes} | \text{sunny, cool, high, strong}) =$

$$v_{NB}(\text{yes}) = P(\text{yes}) P(\text{sunny} | \text{yes}) P(\text{cool} | \text{yes}) P(\text{high} | \text{yes}) P(\text{strong} | \text{yes})$$

$$= \frac{9}{14} * \frac{2}{9} * \frac{3}{9} * \frac{3}{9} * \frac{3}{9}$$

$$= 0.005$$

$$v_{NB}(\text{no}) = P(\text{no})$$

$$P(\text{no}) P(\text{sunny} | \text{no}) P(\text{cool} | \text{no})$$

$$P(\text{no} | \text{sunny, cool, high, strong}) P(\text{high} | \text{no}) P(\text{strong} | \text{no})$$

$$= \frac{5}{14} * \frac{3}{5} * \frac{1}{5} * \frac{4}{5} * \frac{3}{5}$$

$$= 0.021$$

Normalize the probabilities,

$$v_{NB}(\text{yes}) = \frac{v_{NB}(\text{yes})}{v_{NB}(\text{yes}) + v_{NB}(\text{no})}$$

$$= \frac{0.005}{0.005 + 0.021}$$

$$= 0.192$$

$$v_{NB}(\text{no}) = \frac{v_{NB}(\text{no})}{v_{NB}(\text{yes}) + v_{NB}(\text{no})}$$

$$= \frac{0.021}{0.005 + 0.021}$$

$$= 0.808$$